Hill Climbing Algorithm

Hill climbing is a heuristic search algorithm which continuously moves in the direction of increasing height/value to find the peak of the mountain or the best solution to the problem. It terminates when it reaches a peak value where no neighbor has a higher value.

Properties of Hill Climbing Algorithm:

1. **Local search**: It works in problems where knowledge of the entire search space is not available. It only explores the local search space.
2. **Greedy approach**: It only moves in the direction which immediately optimizes the cost/heuristic.
3. **No backtracking**: It cannot backtrack the search space, as it does not remember the previous states.

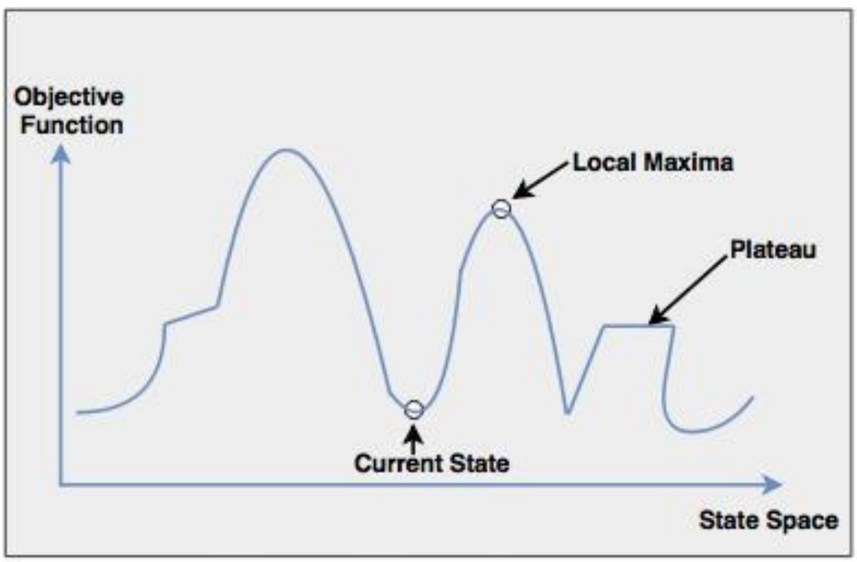

Fig: Hill Climbing : Local Search

Steps:

1. Start with an initial solution (a random point)
2. Evaluate its neighboring solutions
3. If any neighbor is better (low cost/heuristic), move to that one
4. Repeat steps 2-3
5. Stop when no better neighbor is found (local maxima/minima reached)

Drawbacks of Hill Climbing Algorithm:

1. **Local Maximum/Minimum:** The algorithm can get stuck at a peak that seems like the best solution locally, but is not the best globally.

2. **Plateau:** The algorithm may fail to progress in a flat area in the search space where the neighboring solutions have the same value.

3. **Ridge:** The algorithm may fail to follow the correct direction along the ridge that rises in one direction while remaining flat in others.

8 Puzzle Problem with Hill Climbing Algorithm: